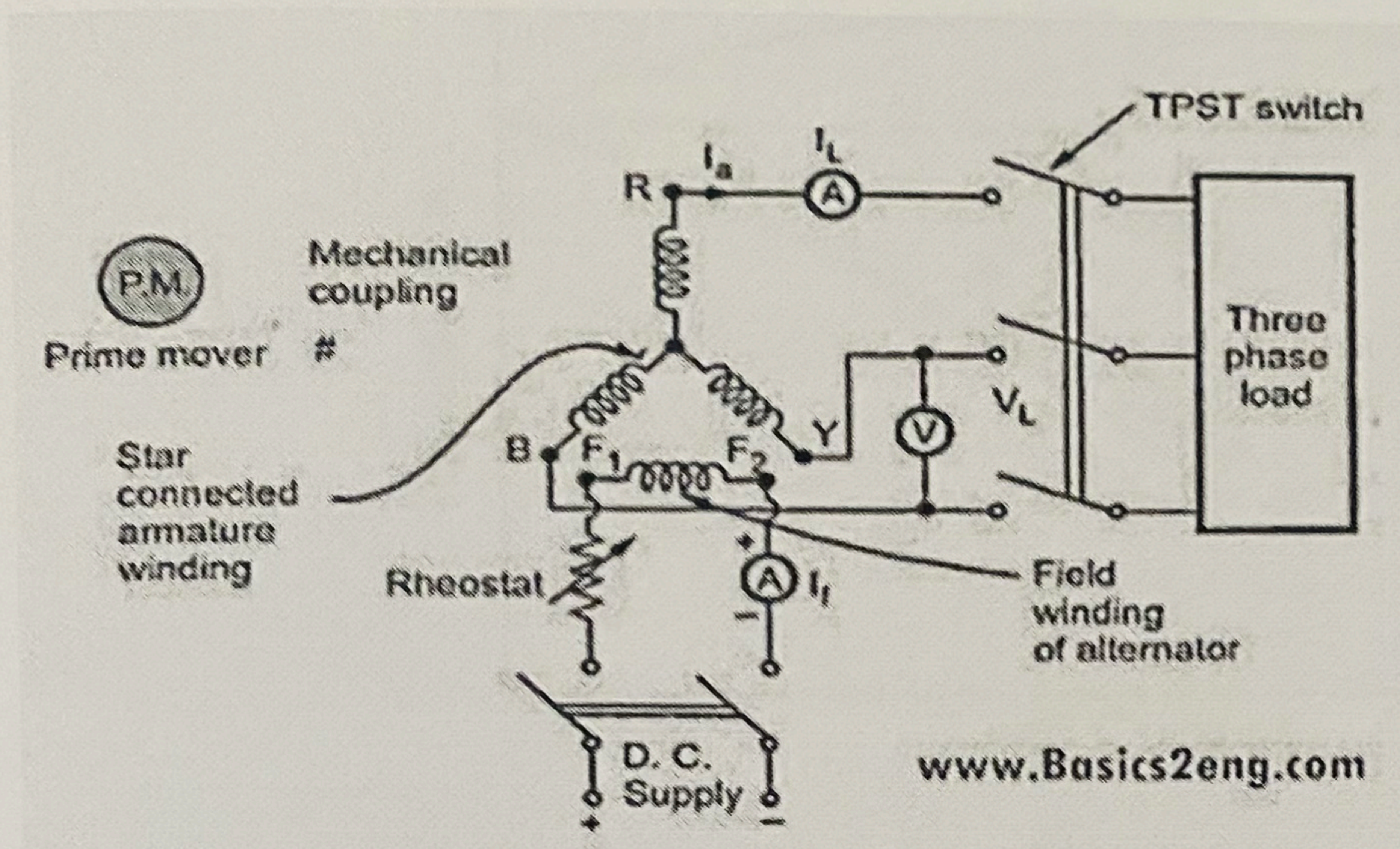




Circuit Diagram



Procedure:-

- Connect the circuit as per the circuit diagram.
- Keep the field rheostat of d.c. motor at minimum position and field rheostat of alternator at maximum position.
- Switch on d.c. supply and start d.c. shunt motor.
- Adjust the speed of d.c. motor to 1500 r.p.m. Adjust rheostat in field circuit of alternator such that it gives rated voltage i.e. 400V. Take down voltmeter reading. This is no load voltage of alternator.
- Switch ON load such that ammeter of each phase indicate same reading.
- Check speed of motor, if it is reduced, adjust the speed to 1500 r.p.m. at this loading condition. • Take down readings of ammeter and voltmeter by gradually increasing load. At each loading condition, speed should be adjusted to 1500 rpm.
- Switch off load bank
- Adjust rheostat of alternator field to maximum position and switch off d.c. supply.
- Plot the graph.

Observation Table:-

Sl. No	I_a amps	V_L (volts)	N rpm
1			
2			
3			
4			
5			